

CASE STUDIES

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Section:1[Practical]

Task 6: File Permissions Using chmod

Objective: Understand Linux file permissions.

```

haasini@Haasini-LAPTOP:~$ cd linux_task4
haasini@Haasini-LAPTOP:~/linux_task4$ touch public.txt private.txt script.sh
haasini@Haasini-LAPTOP:~/linux_task4$ ls
private.txt  public.txt  script.sh
haasini@Haasini-LAPTOP:~/linux_task4$ echo "This is a public file." > public.txt
haasini@Haasini-LAPTOP:~/linux_task4$ echo "This is a private file." > private.txt
haasini@Haasini-LAPTOP:~/linux_task4$ echo "echo Hello from script" > script.sh
haasini@Haasini-LAPTOP:~/linux_task4$ ls -l
total 12
-rw-r--r-- 1 haasini haasini 24 Aug 10 06:10 private.txt
-rw-r--r-- 1 haasini haasini 23 Aug 10 06:10 public.txt
-rw-r--r-- 1 haasini haasini 23 Aug 10 06:10 script.sh
haasini@Haasini-LAPTOP:~/linux_task4$ chmod 600 private.txt
haasini@Haasini-LAPTOP:~/linux_task4$ ls -l private.txt
-rw----- 1 haasini haasini 24 Aug 10 06:10 private.txt
haasini@Haasini-LAPTOP:~/linux_task4$ chmod 644 public.txt
haasini@Haasini-LAPTOP:~/linux_task4$ ls -l public.txt
-rw-r--r-- 1 haasini haasini 23 Aug 10 06:10 public.txt
haasini@Haasini-LAPTOP:~/linux_task4$ chmod +x script.sh
haasini@Haasini-LAPTOP:~/linux_task4$ ls -l script.sh
-rwxr-xr-x 1 haasini haasini 23 Aug 10 06:10 script.sh
haasini@Haasini-LAPTOP:~/linux_task4$ ls -l
total 12
-rw----- 1 haasini haasini 24 Aug 10 06:10 private.txt
-rw-r--r-- 1 haasini haasini 23 Aug 10 06:10 public.txt
-rwxr-xr-x 1 haasini haasini 23 Aug 10 06:10 script.sh
haasini@Haasini-LAPTOP:~/linux_task4$ chmod -x script.sh
haasini@Haasini-LAPTOP:~/linux_task4$ ls -l script.sh
-rw-r--r-- 1 haasini haasini 23 Aug 10 06:10 script.sh
haasini@Haasini-LAPTOP:~/linux_task4$

```

Task 7: File Permission Challenge

Objective: Understand how permissions affect files and directories.

```

haasini@Haasini-LAPTOP:~/permission_lab$ ls -l
total 0
-rw-r--r-- 1 haasini haasini 0 Aug 10 06:18 test.txt
haasini@Haasini-LAPTOP:~/permission_lab$ echo "This is a permission test file." > test.txt
haasini@Haasini-LAPTOP:~/permission_lab$ cat test.txt
This is a permission test file.
haasini@Haasini-LAPTOP:~/permission_lab$ chmod u-r test.txt
haasini@Haasini-LAPTOP:~/permission_lab$ ls -l test.txt
--w-r--r-- 1 haasini haasini 32 Aug 10 06:19 test.txt
haasini@Haasini-LAPTOP:~/permission_lab$ cat test.txt
cat: test.txt: Permission denied
haasini@Haasini-LAPTOP:~/permission_lab$ chmod u+r test.txt
haasini@Haasini-LAPTOP:~/permission_lab$ cat test.txt
This is a permission test file.
haasini@Haasini-LAPTOP:~/permission_lab$ chmod u-w test.txt
haasini@Haasini-LAPTOP:~/permission_lab$ ls -l test.txt
-r--r--r-- 1 haasini haasini 32 Aug 10 06:19 test.txt
haasini@Haasini-LAPTOP:~/permission_lab$ echo "New line" >> test.txt
-bash: test.txt: Permission denied
haasini@Haasini-LAPTOP:~/permission_lab$ chmod u+w test.txt
haasini@Haasini-LAPTOP:~/permission_lab$ echo "New line" >> test.txt
haasini@Haasini-LAPTOP:~/permission_lab$ cat test.txt
This is a permission test file.
New line
haasini@Haasini-LAPTOP:~/permission_lab$ chmod u-x test.txt
haasini@Haasini-LAPTOP:~/permission_lab$ ls -l test.txt
-rw-r--r-- 1 haasini haasini 41 Aug 10 06:20 test.txt
haasini@Haasini-LAPTOP:~/permission_lab$ ./test.txt
-bash: ./test.txt: Permission denied
haasini@Haasini-LAPTOP:~/permission_lab$ |

```

Task 8: Backup and Restore

Objective: Create backups and restore files using cp

```

important.txt linux_task2 linux_task4 organization permission_lab
haasini@Haasini-LAPTOP:~$ mv important.txt linux_task6/
haasini@Haasini-LAPTOP:~$ cd linux_task6
haasini@Haasini-LAPTOP:~/linux_task6$ ls
important.txt
haasini@Haasini-LAPTOP:~/linux_task6$ nano important.txt
haasini@Haasini-LAPTOP:~/linux_task6$ mkdir backup
haasini@Haasini-LAPTOP:~/linux_task6$ ls
backup important.txt
haasini@Haasini-LAPTOP:~/linux_task6$ cp important.txt backup/
haasini@Haasini-LAPTOP:~/linux_task6$ ls backup
important.txt
haasini@Haasini-LAPTOP:~/linux_task6$ mv backup/important.txt backup/important_backup.txt
haasini@Haasini-LAPTOP:~/linux_task6$ ls backup
important_backup.txt
haasini@Haasini-LAPTOP:~/linux_task6$ rm important.txt
haasini@Haasini-LAPTOP:~/linux_task6$ ls
backup
haasini@Haasini-LAPTOP:~/linux_task6$ cp backup/important_backup.txt .
haasini@Haasini-LAPTOP:~/linux_task6$ ls
backup important_backup.txt
haasini@Haasini-LAPTOP:~/linux_task6$ mv important_backup.txt important_restored.txt
haasini@Haasini-LAPTOP:~/linux_task6$ ls
backup important_restored.txt
haasini@Haasini-LAPTOP:~/linux_task6$ cat important_restored.txt
This is important informatiion.
Linux is an operating systemc.
Files can be backed up.
Backup protects important data.
The cp command copies files.
The mv command moves files.
The rm command deletes files.
Restoring data is useful.
Linux file management is easy.
This is my backup practice.
haasini@Haasini-LAPTOP:~/linux_task6$ |

```

Task 9: Complete Linux File Management Challenge

Objective: Use almost all the commands together.

```
haasini@Haasini-LAPTOP:~$ mkdir linux_project
haasini@Haasini-LAPTOP:~$ cd linux_project
haasini@Haasini-LAPTOP:~/linux_project$ pwd
/home/haasini/linux_project
haasini@Haasini-LAPTOP:~/linux_project$ mkdir source backup docs
haasini@Haasini-LAPTOP:~/linux_project$ ls
backup  docs  source
haasini@Haasini-LAPTOP:~/linux_project$ touch source/main.c source/helper.c source/README.txt
haasini@Haasini-LAPTOP:~/linux_project$ ls source
README.txt  helper.c  main.c
haasini@Haasini-LAPTOP:~/linux_project$ nano source/README.txt
haasini@Haasini-LAPTOP:~/linux_project$ head -n 5 source/README.txt
Linux is an operating system.
Linux supports file management.
Directories organize files.
The cp command copies files.
The mv command moves files.
haasini@Haasini-LAPTOP:~/linux_project$ tail -n 5 source/README.txt
C is a programming language.
Source files contain program code.
Backup protects important files.
Documentation explains a project.
This is my Linux project.
haasini@Haasini-LAPTOP:~/linux_project$ tail -n 5 source/README.txt
C is a programming language.
Source files contain program code.
Backup protects important files.
Documentation explains a project.
This is my Linux project.
haasini@Haasini-LAPTOP:~/linux_project$ cp source/main.c backup/
haasini@Haasini-LAPTOP:~/linux_project$ ls backup
main.c
haasini@Haasini-LAPTOP:~/linux_project$ cp source/README.txt docs/
haasini@Haasini-LAPTOP:~/linux_project$ ls docs
README.txt
haasini@Haasini-LAPTOP:~/linux_project$ |
```



```

haasini@Haasini-LAPTOP:~/linux_project$ ls docs
README.txt
haasini@Haasini-LAPTOP:~/linux_project$ mv source/helper.c source/functions.c
haasini@Haasini-LAPTOP:~/linux_project$ ls source
README.txt  functions.c  main.c
haasini@Haasini-LAPTOP:~/linux_project$ chmod 644 source/README.txt
haasini@Haasini-LAPTOP:~/linux_project$ ls -l source/README.txt
-rw-r--r-- 1 haasini haasini 672 Aug 10 06:53 source/README.txt
haasini@Haasini-LAPTOP:~/linux_project$ rm backup/main.c
haasini@Haasini-LAPTOP:~/linux_project$ ls backup
haasini@Haasini-LAPTOP:~/linux_project$ rmdir backup
haasini@Haasini-LAPTOP:~/linux_project$ ls
docs  source
haasini@Haasini-LAPTOP:~/linux_project$ ls -R
.:
docs  source

./docs:
README.txt

./source:
README.txt  functions.c  main.c
haasini@Haasini-LAPTOP:~/linux_project$ tree
.
├── docs
│   └── README.txt
└── source
    ├── README.txt
    ├── functions.c
    └── main.c

3 directories, 4 files
haasini@Haasini-LAPTOP:~/linux_project$ |

```

Task 10: Mini Linux Administration Challenge

Objective: Complete a real-world-style file management exercise.

```
haasini@Haasini-LAPTOP:~/student_portal$ mkdir student_portal
haasini@Haasini-LAPTOP:~/student_portal$ cd student_portal
haasini@Haasini-LAPTOP:~/student_portal$ pwd
/home/haasini/student_portal
haasini@Haasini-LAPTOP:~/student_portal$ mkdir src docs backup logs config
haasini@Haasini-LAPTOP:~/student_portal$ ls
backup  config  docs  logs  src
haasini@Haasini-LAPTOP:~/student_portal$ touch src/main.c src/student.c docs/README.txt config/config.txt logs/system.log
haasini@Haasini-LAPTOP:~/student_portal$ ls -R
.:
backup  config  docs  logs  src

./backup:

./bonfug:
config.txt

./docs:
README.txt

./logs:
system.log

./src:
main.c  student.c
haasini@Haasini-LAPTOP:~/student_portal$ nano logs/system.log
haasini@Haasini-LAPTOP:~/student_portal$ head -n 5 logs/system.log
System started successfully.
Student portal initialized.
Database connection established.
User login service started.
Student records loaded.
haasini@Haasini-LAPTOP:~/student_portal$ tail -n 5 logs/system.log
Profile updated successfully.
Student searched successfully.
Report generated.
Backup process completed.
System running normally.
haasini@Haasini-LAPTOP:~/student_portal$ cp docs/README.txt backup/
haasini@Haasini-LAPTOP:~/student_portal$ ls backup
README.txt
haasini@Haasini-LAPTOP:~/student_portal$ |
```

```
haasini@Haasini-LAPTOP:~/student_portal$ ls backup
README_backup.txt
haasini@Haasini-LAPTOP:~/student_portal$ chmod 600 config/config.txt
haasini@Haasini-LAPTOP:~/student_portal$ ls -l config/config.txt
-rw----- 1 haasini haasini 0 Aug 10 07:00 config/config.txt
haasini@Haasini-LAPTOP:~/student_portal$ chmod 644 docs/README.txt
haasini@Haasini-LAPTOP:~/student_portal$ ls -l docs/README.txt
-rw-r--r-- 1 haasini haasini 0 Aug 10 07:00 docs/README.txt
haasini@Haasini-LAPTOP:~/student_portal$ rm backup/README_backup.txt
haasini@Haasini-LAPTOP:~/student_portal$ ls backup
haasini@Haasini-LAPTOP:~/student_portal$ rmdir backup
haasini@Haasini-LAPTOP:~/student_portal$ ls
config docs logs src
haasini@Haasini-LAPTOP:~/student_portal$ ls -R
.
config docs logs src

./config:
config.txt

./docs:
README.txt

./logs:
system.log
./src:
main.c student.c
haasini@Haasini-LAPTOP:~/student_portal$ tree
.
├── config
│   └── config.txt
├── docs
│   └── README.txt
├── logs
│   └── system.log
└── src
    ├── main.c
    └── student.c

5 directories, 5 files
haasini@Haasini-LAPTOP:~/student_portal$ |
```